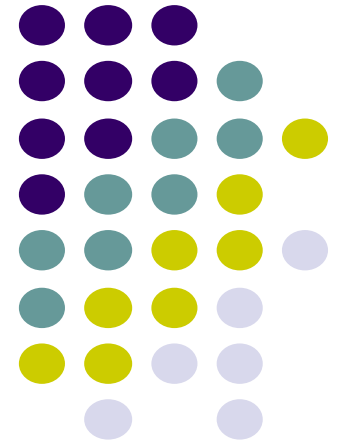


ErSE222: Machine learning in Geoscience

Feb. 25th, 2026

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SGD with momentum

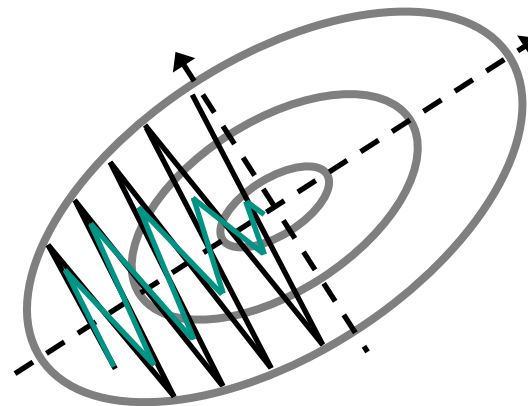
Idea of momentum (from Polyak and Nesterov in 60'): use update that is an exponentially decaying moving average of the past gradients.

Velocity Momentum term $[0,1)$

↓ ↙

$$\mathbf{v}_{i+1} = \gamma \mathbf{v}_i - \mathbf{g}_{i+1} = \gamma \mathbf{v}_i - \frac{\alpha}{N_b} \sum_{j=1}^{N_b} \nabla \mathcal{L}_j$$

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i + \mathbf{v}_{i+1}$$





Adaptive Gradient (AdaGrad)

Idea of AdaGrad : scale gradients by inverse of square root of sum of historical squares gradients

$$\mathbf{g}_{i+1} = \frac{1}{N_b} \sum_{j=1}^{N_b} \nabla \mathcal{L}_j$$

$$\mathbf{r}_{i+1} = \mathbf{r}_i + \mathbf{g}_{i+1} \cdot \mathbf{g}_{i+1}$$

$$\Delta \boldsymbol{\theta}_{i+1} = -\frac{\alpha}{\delta + \sqrt{\mathbf{r}_{i+1}}} \cdot \mathbf{g}_{i+1}$$

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i + \Delta \boldsymbol{\theta}_{i+1}$$

Parameter dependent rescaling



Root Mean Square Propagation (RMSProp)

Idea of RMSProp : gradient accumulation is *exponentially damped*

$$\mathbf{g}_{i+1} = \frac{1}{N_b} \sum_{j=1}^{N_b} \nabla \mathcal{L}_j$$

Slower decay of LR

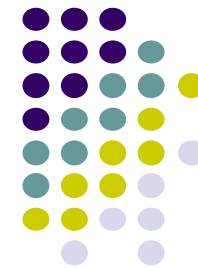
$$\mathbf{r}_{i+1} = \rho \mathbf{r}_i + (1 - \rho) \mathbf{g}_{i+1} \cdot \mathbf{g}_{i+1}$$

$$\Delta \boldsymbol{\theta}_{i+1} = -\frac{\alpha}{\delta + \sqrt{\mathbf{r}_{i+1}}} \cdot \mathbf{g}_{i+1}$$

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i + \Delta \boldsymbol{\theta}_{i+1}$$

Parameter dependent rescaling

Adam = Momentum + RMSProp



$$g_t = \frac{1}{N_b} \sum_{j=1}^{N_b} \nabla \mathcal{L}_j$$

$$v_t = \rho_1 v_{t-1} + (1 - \rho_1) g_t \quad (\text{velocity term})$$

$$r_t = \rho_2 r_{t-1} + (1 - \rho_2) g_t^2 \quad (\text{scaling term})$$

$$\hat{v}_t = \frac{v_t}{1 - \rho_1^t} \quad (\text{bias correction})$$

$$\hat{r}_t = \frac{r_t}{1 - \rho_2^t} \quad (\text{bias correction})$$

$$\Delta \theta_t = - \frac{\alpha}{\delta + \sqrt{\hat{r}_t}} \hat{v}_t$$

$$\theta_t = \theta_{t-1} + \Delta \theta_t$$

$$\alpha = 0.001 \quad \rho_1 = 0.9 \quad \rho_2 = 0.999 \quad \delta = 10^{-8}$$



Concepts and Definitions

Skip connections



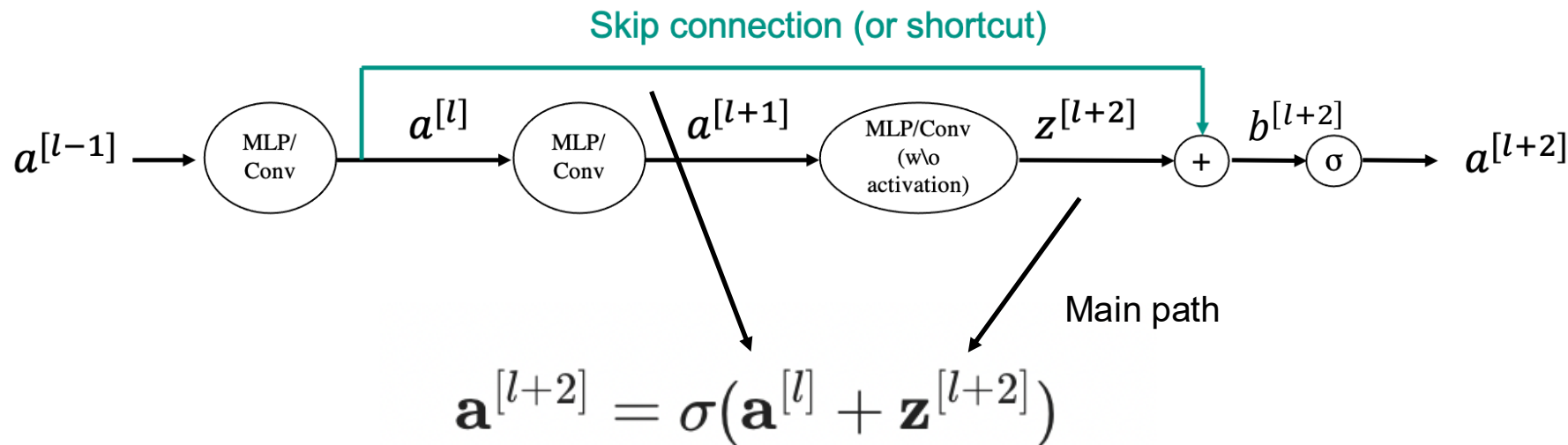
Deep neural networks can learn more complex relationships...

... but they are prone to slow training due to **vanishing gradients**

Skip connections: introduced in 2015 under the name of **Residual blocks**

take the activation of a certain layer and feed it directly to another layer further down in the computational graph.

Skip connections



Why successful?

$$\mathbf{b}^{[l+2]} = \mathbf{a}^{[l]} + \mathbf{z}^{[l+2]} = \mathbf{a}^{[l]} + f_{l+2}(f_{l+1}(\mathbf{a}^{[l]}))$$

$$\mathbf{a}^{[l+2]} = \sigma(\mathbf{b}^{[l+2]})$$

$$\frac{\partial \mathbf{b}^{[l+2]}}{\partial \mathbf{a}^{[l]}} = 1 + \frac{\partial \mathbf{a}^{[l+1]}}{\partial \mathbf{a}^{[l]}} \frac{\partial \mathbf{z}^{[l+2]}}{\partial \mathbf{a}^{[l+1]}}$$

ensures that gradient is never 0!

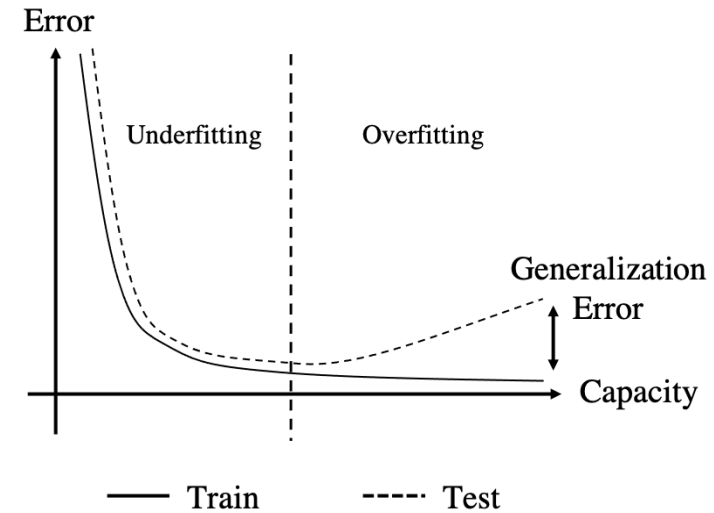
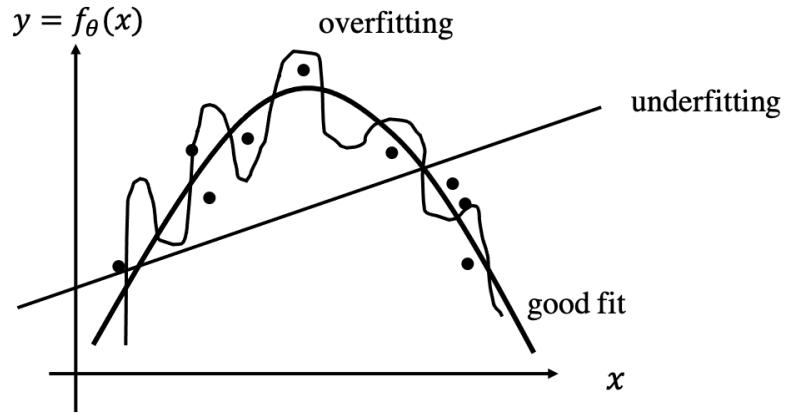
ML performance



Two different (usually contrasting) goals:

- **Bias:** ability to produce a small training error. When the error is small, we say that we have a model with low bias. Vice versa, when the error is large, the model have high bias. In this case, the model is said to be *underfitting* the data;
- **Variance:** ability to produce comparable training and testing error. In other words, if we define *gap* to be the difference between the testing and training errors, this is also the ability to produce a small gap. When the gap is large, the model is said to be *overfitting* the data.

ML performance

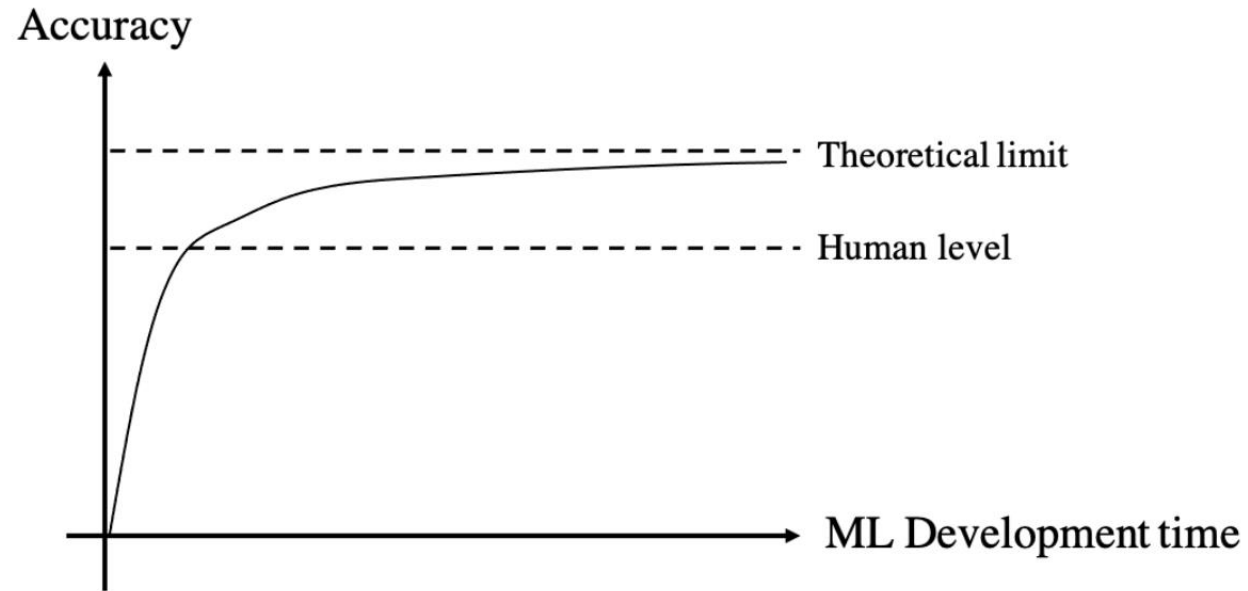


Capacity = size of the model

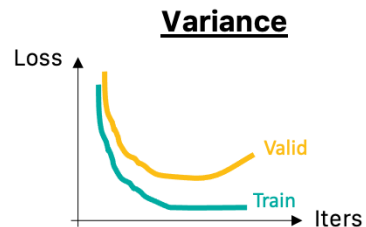
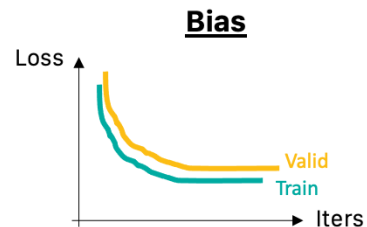
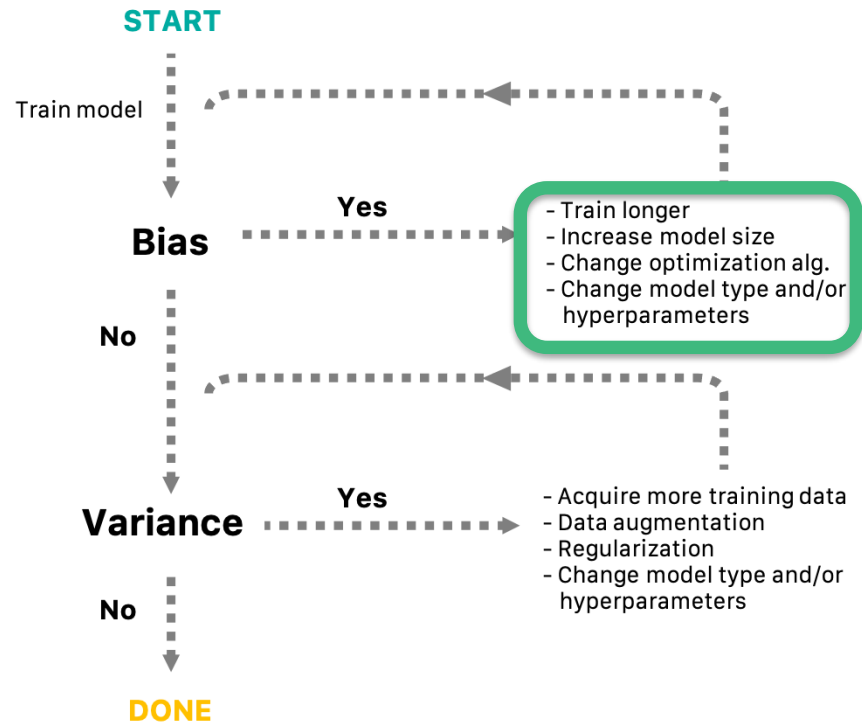
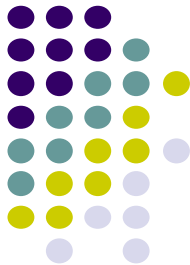
Expectations from ML model



Always set expectations ahead of training!



Summary



ML hyperparameters



Hyperparameter: *parameter of a learning algorithm that is **not learned**, but chosen upfront*

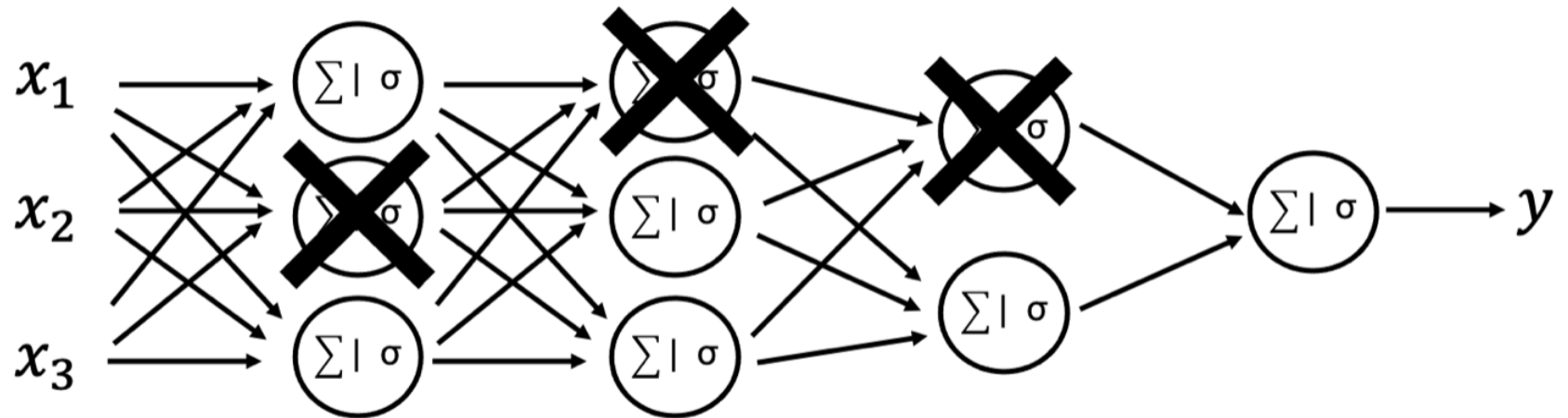
- model size (capacity)
- activation function;
- optimizer,
- learning rate
- additional optimizer hyperparameters,
- batch size...

Early Stopping



It involves stopping the training process before the model fully converges on the training data, preventing overfitting.

Regularization - dropout



Data augmentation



Availability of large set of trustworthy training data = key to success of ML

In the presence of few training data, apply transformations to enrich the dataset



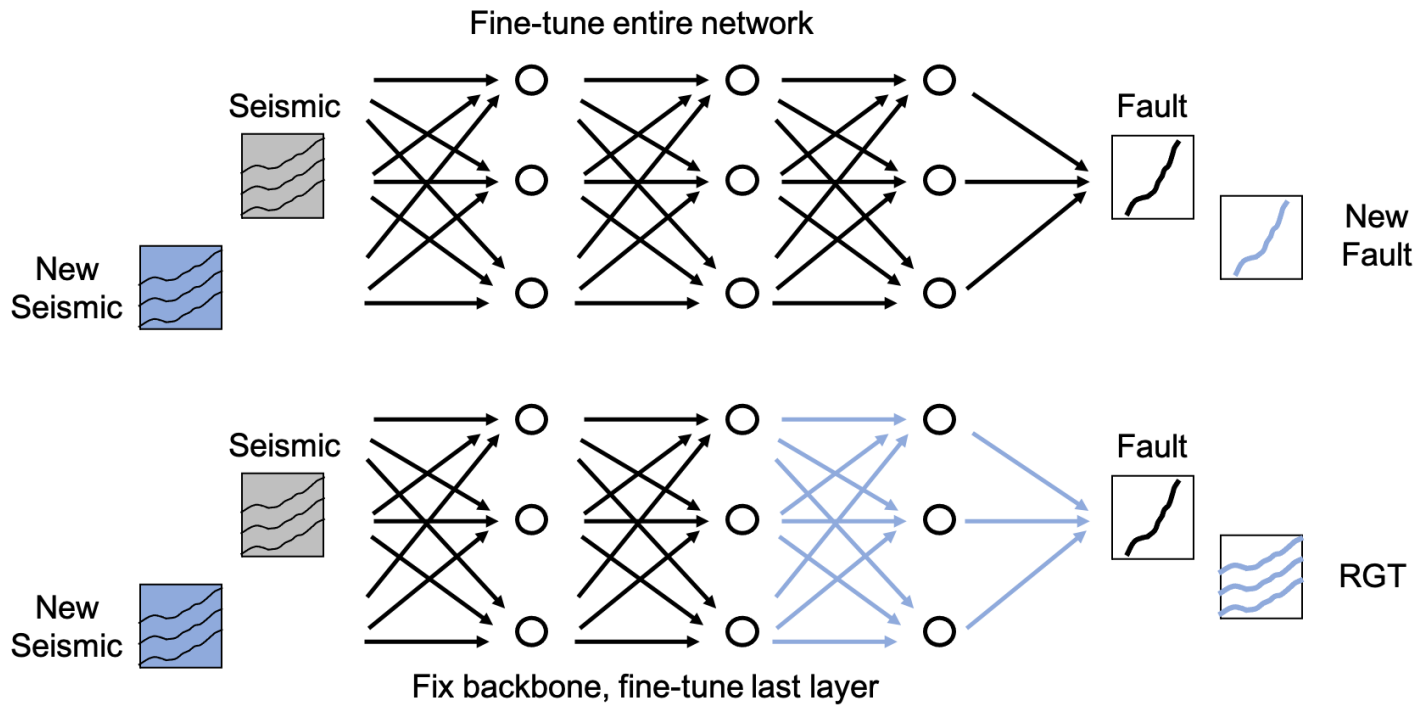
Ex: images

- cropping,
- flipping,
- rotating,
- shearing,
- averaging,
- color shifting

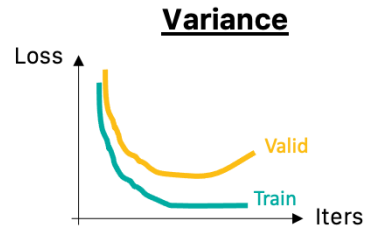
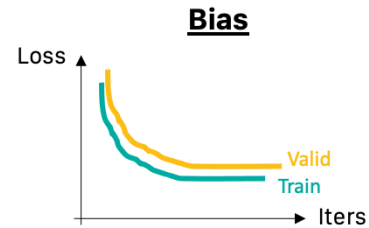
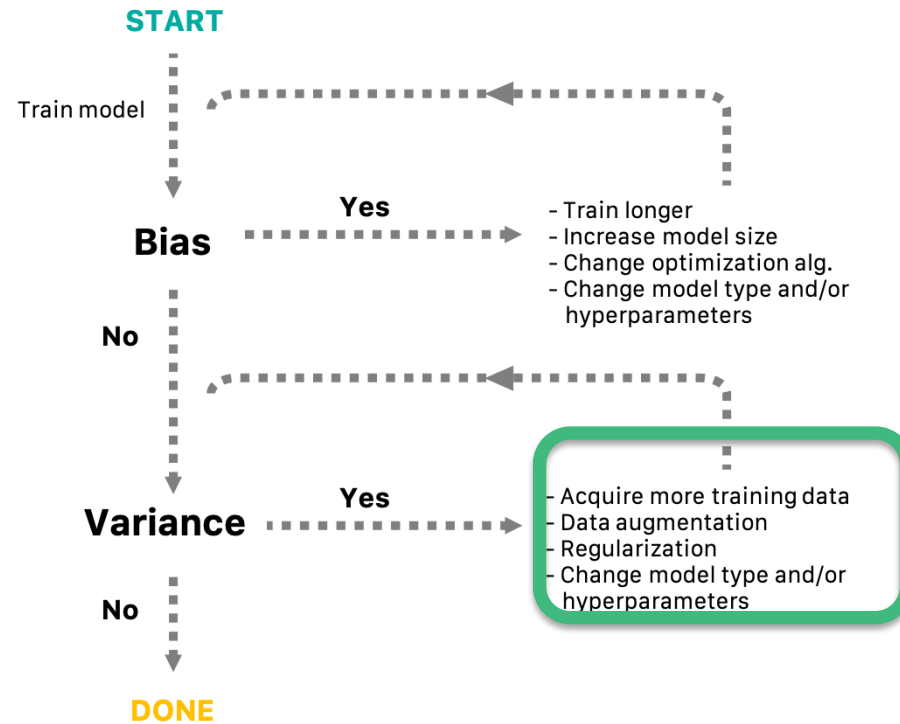
Transfer learning



Special case of data augmentation: use model trained in other data



Summary





Thank You

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